

Stage 3: From direct regression to draft-to-refine

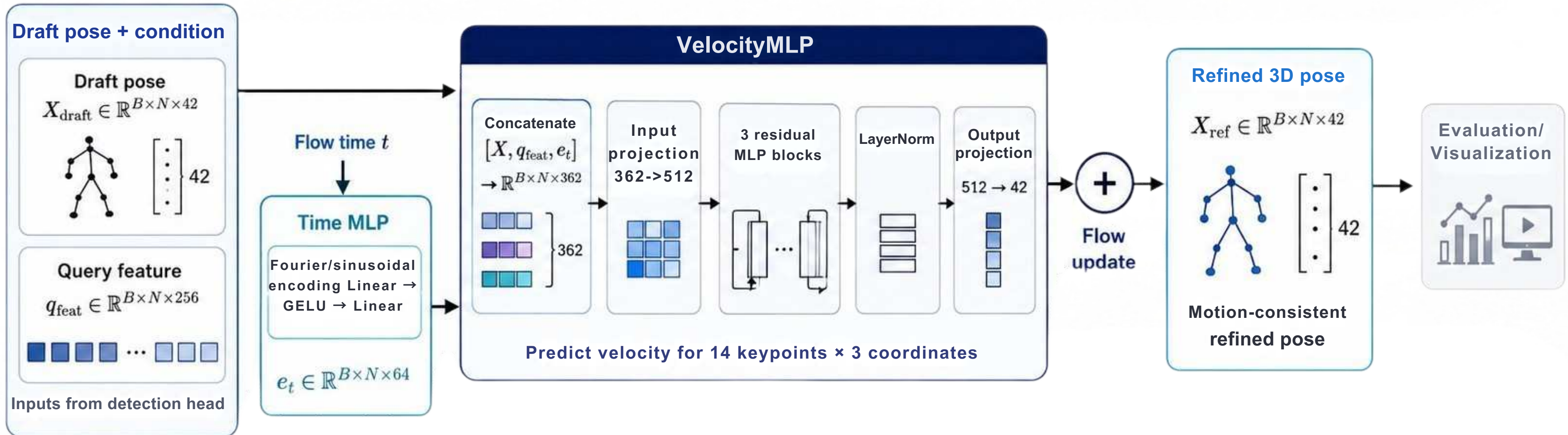
Rectified-Flow Architecture

$$\hat{X}_{\text{ref}} = \text{Flow}(X_{\text{draft}}, q_{\text{feat}}, t), \quad v_{\theta} = f_{\theta}(X_t, q_{\text{feat}}, t)$$

Rectified Flow update

$$X_{t+\Delta t} = X_t + \Delta t v_{\theta}(X_t, q_{\text{feat}}, t)$$

Current pose + predicted velocity refined pose



Residual pose refinement with Rectified Flow